Cpts 317. 10/12.

Emptiness:

Given: a FA M

Q. Alm: $L(M) = \emptyset$?

Alg: input M

" $L(M) \neq \emptyset$ iff there is a word w that is accepted by M

" iff there is a walk from initial in M to accepting in M on which the word is w, for some w

" iff there is a walk from initial to accepting.

Run DFS on the graph of M
and see if initial reaches accepting.

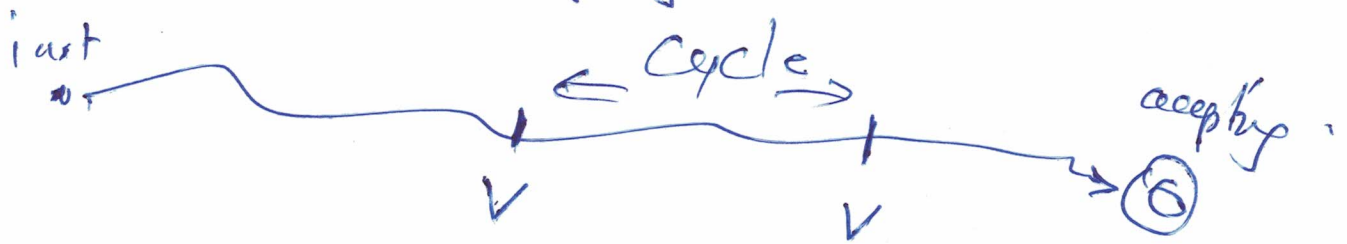
Infinitness.

Given: a FA M

Question: L(M) is an inf. language?

Alg: input M

L(M) is inf. iff there is a walk from initial to accepting where there is a cycle.

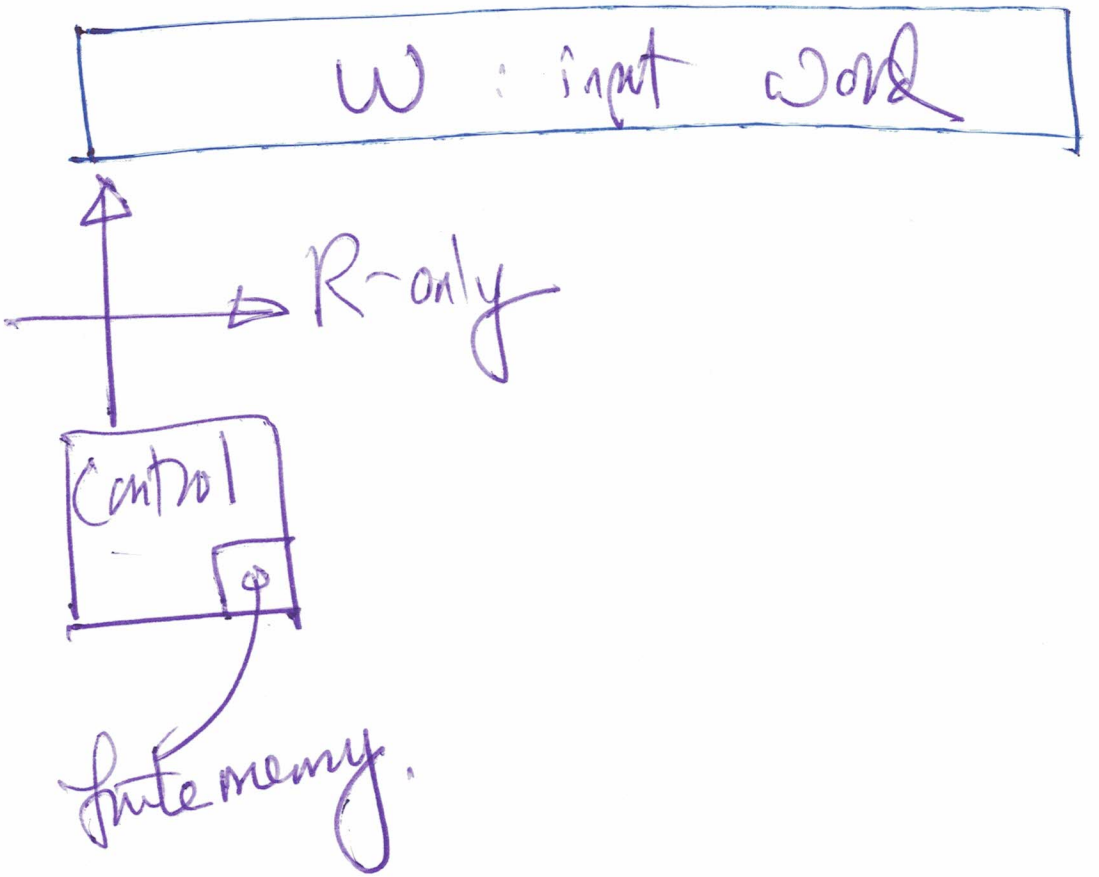


Run an alg to detect whether there is a cycle on a walk from init to accepting.

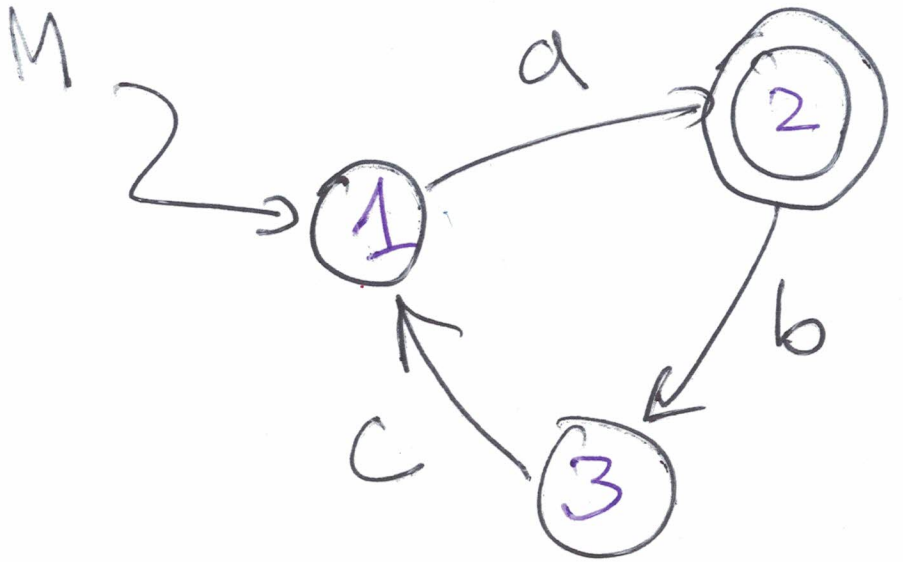
Tarjan alg (cpts 350 next semester).

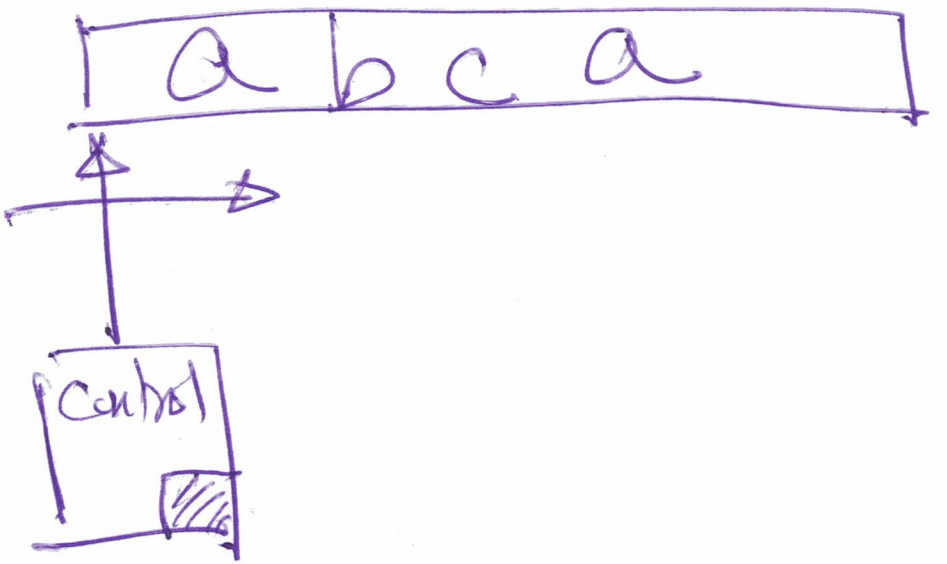
(Forget about structural induction) new ways to do proofs.

FA can be understood as:



Exaple.





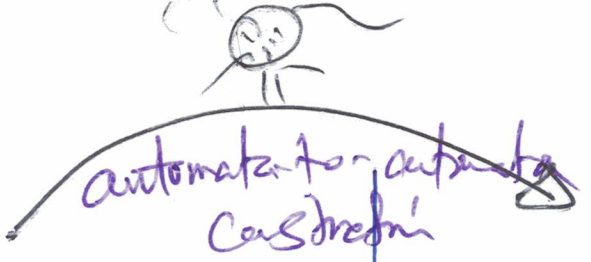
Note on NFA:

as an NFA M, M can "guess" next state;

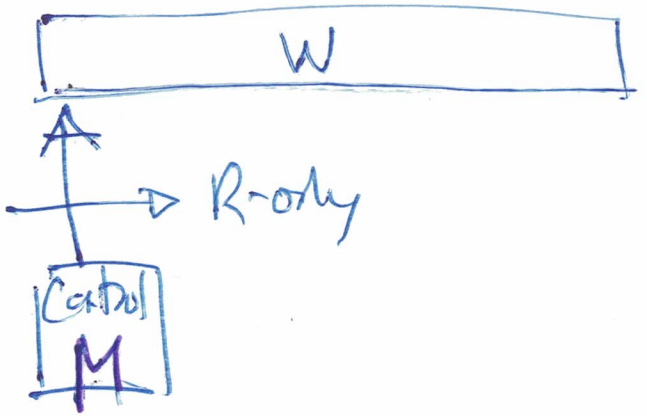
Thm. if L is reg, then End(L, b) is reg.

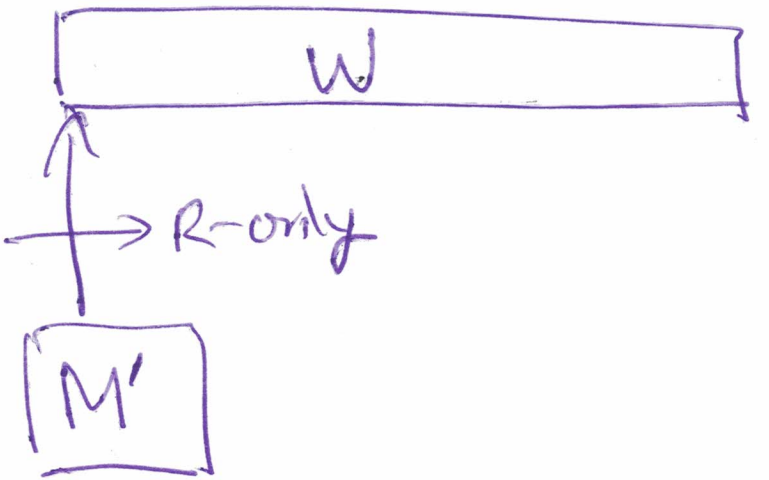
Proof. Since L is reg, we have a FA M to accept L. Now, we will construct, from M, a new FA M' to accept End(L, b). M' runs as follows. On an input word w, M' faithfully simulates M running on the w,

M



M'





M says yes on w iff $w \in L$.

M' says on w iff
$w \in End(L, b)$
iff $w \in L$ and w ended with b
iff M says yes on w and w is ended with b.

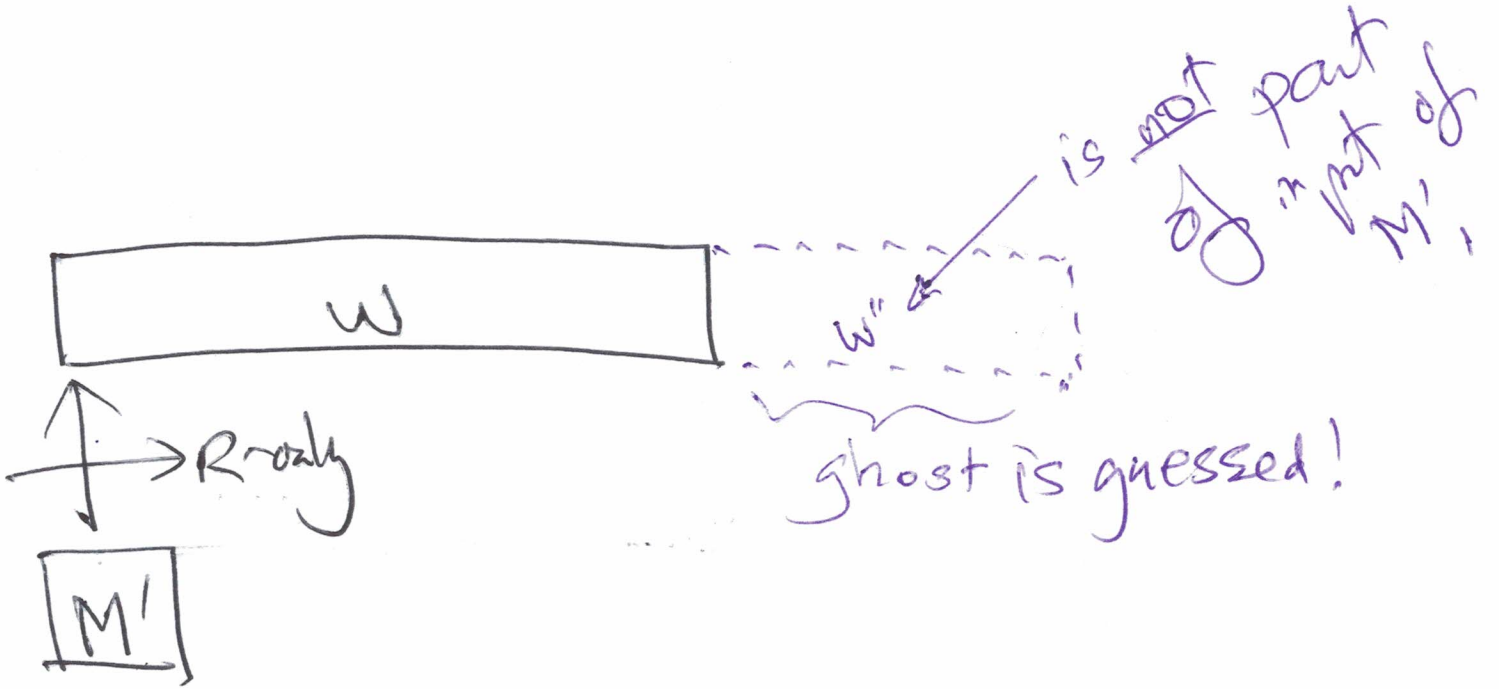
At the end of input ω, M' says yes if M says yes and the ending letter is b. Clearly, M' accepts $End(L, b)$ and the result follows. $\blacksquare$

Harder Example.

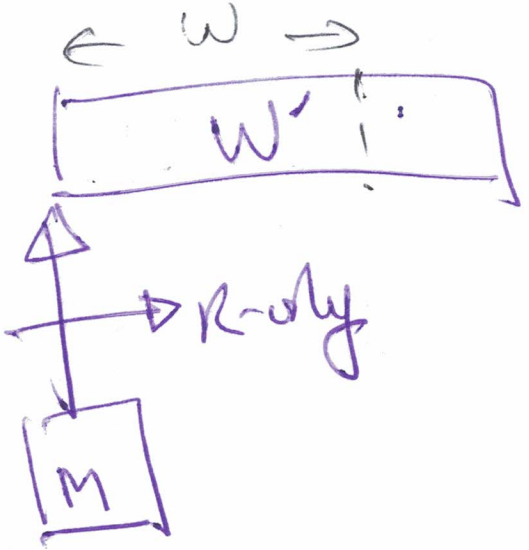
Thm. if L is reg, then so is

$$\text{Prefix}(L) = \{ w : \exists w' \in L,\ w \prec w' \}.$$

Proof.

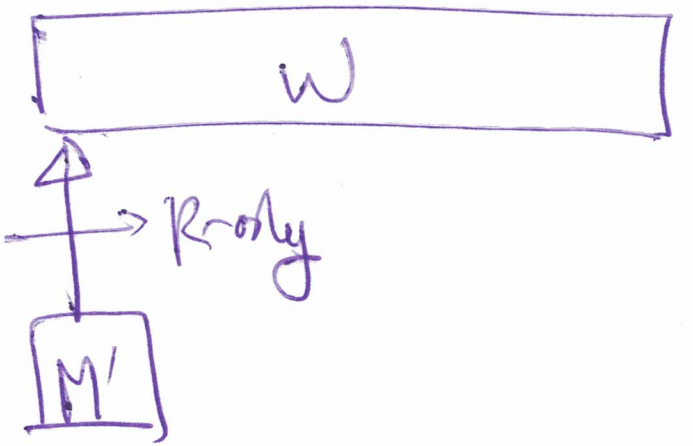


M



M says yes on w'
iff $w' \in L$.

M'



M' says yes on w
iff $w \in \text{Prefix}(L)$
iff $\exists w' \in L$, w is a prefix of w'.
iff $\exists w'$, M says yes on w', w is a prefix of w'.

(Difficulty: w' is <u>not</u> given!

w is a prefix of w').

Proof: Since L is reg, we have a FA M to accept L. Now we construct a FA M' to accept Prefix(L) as follows. On input w, M' simulates M faithfully. At the end of input w, M' guesses, symbol by symbol, a word w'' and let M continue to run on the guessed word. At the end of the guessed word, if M says yes, then M' says yes on w. Clearly, M' accepts Prefix(L). Result follows. ∎